

MATHEMATICS MOCK TEST

UC02 Percentages — Mock Paper for the Test on Tuesday 13 October 2026

Name: _____ Class: _____ Date: _____ Score: _____ / 100

Time allowed: 40 minutes | Total: 100 marks

Answer **ALL** questions. Show your working clearly in the space provided. Calculators are **NOT** allowed.

Write conversions as chains (e.g. $5/8 \rightarrow 0.625 \rightarrow 62.5\%$). Every word problem must end with a $\therefore$ conclusion sentence.

Section A: Fraction, Decimal and Percentage Conversion (18 marks)

1. Complete the following table. Write your answers as chains. Simplify all fractions to their lowest terms. **(10 marks)**

	Fraction	Decimal	Percentage
(a)	$\frac{5}{8}$		
(b)		0.625	
(c)	$\frac{76}{100}$		
(d)			120%
(e)		0.42	

2. Express each of the following as a percentage. **(8 marks)**

(a) $\frac{13}{20}$

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(b) $1\frac{1}{5}$

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Section B: Ordering

(12 marks)

3. Arrange the following in **ascending order** (smallest to largest). Convert to decimals first.

(8 marks)

(a) $\frac{5}{7}$, 0.64, 71%, $0.\dot{7}\dot{1}$

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(b) $\frac{1}{1000}$, $\frac{11}{1000}$, 0.11%, 0.0108

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4. Arrange the following in **ascending order**.

(4 marks)

0.37, $\frac{2}{5}$, 38%, $0.\dot{3}\dot{9}$

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Section C: Finding a Percentage of a Quantity

(16 marks)

5. Find the value of each of the following.

(12 marks)

(a) 17.5% of 44

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(b) 62.5% of 96

(c) 130% of 45

6. 15 300 of the 17 000 books in a library are story books. What percentage of the books are story books? **(4 marks)**

Section D: Percentage Increase and Decrease

(36 marks)

7. A school hall has 500 seats. This year the number of seats is **16% more** than last year. How many seats are there now? **(6 marks)**

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8. A test has 25 questions. Steve answered **20% of them wrong**. How many questions did he answer correctly? **(6 marks)**

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9. A chair costs \$200. After a **15% price increase**, what is the new price of the chair? **(6 marks)**

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10. Amy has \$250. She spends **30% of it** on books, then pays \$145 for a school bag. How much money does she have left? **(8 marks)**

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11. (a) There are 800 students in a school. 45% of them are boys. How many girls are there? **(5 marks)**

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(b) A jacket costs \$450. After a **10% price increase**, what is the new price of the jacket? **(5 marks)**

Section E: Challenge

(18 marks)

12. (a) A shop sold 72 dresses last week. This week the number of dresses sold was **25% more** than last week. How many dresses were sold this week? **(5 marks)**

(b) A toy costs \$240. During a sale its price is **decreased by 10%**. What is the new price of the toy? Show both the multiplier method and the subtraction method. **(5 marks)**

13. (a) 18 out of 25 students passed a quiz. What percentage of the students passed? **(4 marks)**

(b) A water bottle holds 0.6 L. Its capacity is increased by 50%. How much water can it hold now? **(4 marks)**

— End of Paper —

Answer Key

1. (a) 0.625, 62.5% (b) $\frac{5}{8}$, 62.5% (c) $\frac{19}{25}$, 0.76, 76% (d) $1\frac{1}{5}$, 1.2 (e) $\frac{21}{50}$, 42%
2. (a) 65% (b) 120%
3. (a) $0.64 < 71\% < \frac{5}{7} < 0.\dot{7}1$ (b) $\frac{1}{1000} < 0.11\% < 0.0108 < \frac{11}{1000}$
4. $0.37 < 38\% < 0.\dot{3}9 < \frac{2}{5}$
5. (a) 7.7 (b) 60 (c) 58.5
6. 90%
7. $500 \times 116\% = 500 \times 1.16 = 580$. $\therefore$ There are 580 seats now.
8. $25 \times 20\% = 5$ (wrong); $25 - 5 = 20$. $\therefore$ He answered 20 questions correctly.
9. $200 \times 115\% = 200 \times 1.15 = 230$. $\therefore$ The new price of the chair is \$230.
10. $250 \times 30\% = 75$ (books); $250 - 75 = 175$; $175 - 145 = 30$. $\therefore$ She has \$30 left.
11. (a) $800 \times 45\% = 360$ (boys); $800 - 360 = 440$. $\therefore$ There are 440 girls. (b) $450 \times 110\% = 450 \times 1.1 = 495$. $\therefore$ The new price of the jacket is \$495.
12. (a) $72 \times 125\% = 72 \times 1.25 = 90$. $\therefore$ 90 dresses were sold this week. (b)
Multiplier: $240 \times 90\% = 240 \times 0.9 = 216$. Subtraction: $240 \times 10\% = 24$; $240 - 24 = 216$. $\therefore$ The new price of the toy is \$216.
13. (a) $18 \div 25 \times 100\% = 72\%$ (b) $0.6 \times 150\% = 0.6 \times 1.5 = 0.9$ L. $\therefore$ It can hold 0.9 L now.